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Some objective functions and ideas to optimize experimental designs in artificial selection programs

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Abstract

Experimental designs in artificial selection programs must balance limited resources with the need for accurate genetic evaluation across diverse environments. We propose a decision-point framework that decomposes the overall design process into sequential allocation steps, each optimized using objective functions based on representativeness. Using stochastic simulations, we evaluated how these objective functions influence accuracy and genetic gain across multiple design decisions. Key results emerged. First, the selection of a sufficiently large and representative set of environments had the greatest impact on across target population of environments (TPEs) accuracy. When the TPE was complex, sparse-testing strategies consistently outperformed fully balanced designs, even when total plot numbers were held constant. Second, environment selection using a modified optimal-contribution function ($-\mathbf{q}'\mathrm{D}\mathbf{q}$) outperformed random sampling, *k*-means, and hierarchical clustering, yielding up to a 20% gain in accuracy and reduced uncertainty. Third, once environments were fixed, the allocation of entries across environments had a moderate but meaningful effect, especially when genetic correlations varied widely across the TPE. Fourth, within-environment design decisions strongly influenced accuracy: optimal field size depended on the level of spatial heterogeneity, while replication had relatively limited impact when genomic relationship matrices were used. Finally, optimized spatial allocation improved accuracy under strong spatial noise, though at increased computational cost. Overall, this framework confirms prior findings while offering a practical and computationally efficient approach for optimizing phenotypic evaluations. By tailoring objective functions to specific decision points, breeding programs can improve accuracy, reduce risk, and make better use of limited resources.

Abbreviations: G × E, genotype-by-environment interaction; GRM, genomic relationship matrix; *hclust*, hierarchical clustering; MVN, multivariate normal distribution; p-rep, partially replicated; QTL/QTLs, quantitative trait locus/loci; TPE, target population of environments; TPG, target population of genotype.

Plain Language Summary

The design of experiments is used across a wide variation of science fields to draw valid conclusions from an experiment involving different treatments to test a set of hypotheses. In artificial selection experiments, a proper design of experiments allows to increase the accuracy of individual selection to produce the needed variety in the farms. The aim of this study is to test different objective functions based on quantitative genetics theory to optimize the resources allocated to test the individuals in the different locations but also in plots within the locations. Using simulation, we showed the value of the objective functions proposed to optimally allocate the constrained resources (in terms of plots and entries). Depending on the decision, a different objective function can be used to guide an optimal allocation of entries across and within environments. Different optimization methodologies can be used to fit the objective functions. Greater selection accuracy was achieved when the proposed objective functions were used.

1 | INTRODUCTION

Since the early times of experimental design, the principles of replication, randomization, and blocking have been of extreme value for researchers to ensure valid inferences to answer hypotheses (Box, 1980; Fisher, 1935). In the case of artificial selection programs, the design of experiments traditionally aims to reduce spatial, management, and residual noise in field evaluations to improve the estimation of genetic and breeding values (Bernardo, 2002; Cullis et al., 2006). As geneticists unraveled the importance of the genotype by environment interactions (G × E) in the last decades, experimental designs have focused more and more on partitioning the environmental sources of variation and the main genotype effects out of the G × E and understanding the role of weather and soil information in the observed interactions instead of only addressing within-location sources of variation (Cooper et al., 2023; de Leon et al., 2016; Tolhurst et al., 2022; van Eeuwijk et al., 1996).

Many authors focused on the within-location optimization (Cullis et al., 2006; Hoefler et al., 2020; Qiao et al., 2000; Smith et al., 2006; Zystro et al., 2018) in lieu of more holistic questions to optimize the allocation of resources. In this study, we propose a structured decision-point framework in which each stage of the breeding process is optimized using objective functions tailored to that decision. This partitioning allows programs to approximate a near-optimal allocation without requiring exhaustive joint optimization. The proposed objective functions associated with these sub-processes are based on the concept of representativeness, a principle commonly used in quantitative genetics. The objective functions performance is evaluated using stochastic simulations (see Supporting Information), a proven approach for assessing design and optimization strategies in artificial selection programs (Covarrubias-Pazaran et al., 2022; Gaynor et al., 2021; Pryce & Daetwyler, 2011).

We organize the manuscript around sequential decision points that breeding programs commonly face when allocating resources. These decision points are framed as the following key questions:

1. How many and which environments represent the target population of environments (TPE)?
2. How many and which individuals best represent the target population of genotypes (TPG)?
3. How should entries and environments be optimally paired?
4. How many plots and what level of replication should be used within each environment?
5. How should entries be spatially arranged within an environment?

Although jointly optimizing all decisions would be ideal, the combinatorial complexity makes such an approach impractical. By addressing each decision independently, we obtain solutions that are not globally optimal but sufficiently close to guiding practical breeding decisions. Users with substantial computational resources may still apply the full suite of objective functions simultaneously. It is important to note that the first two questions intersect with the broader issue of defining the TPE. Our aim is not to redefine the TPE itself, but rather to propose objective functions that quantify representativeness when selecting environments within a given TPE.

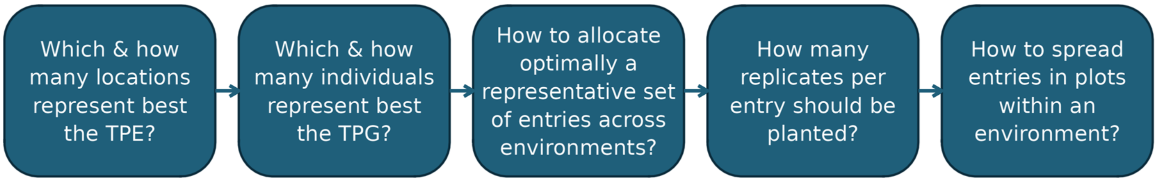

FIGURE 1 Chronological decision-points flowchart. Nodes/boxes represent decisions, and arrows represent the flow of decision points in a chronological order. Each of these boxes uses an objective function to optimize the decision (see Section 2) or a simulated grid of treatments. TPE, target population of environments; TPG, target population of genotypes.

2 | MATERIALS AND METHODS

We structured the methods based on decision-making points that artificial selection programs are usually exposed to. Figure 1 summarizes the chronological decision points. We simulated TPGs and TPEs based on common population structures and dimensions found in small- and medium-size breeding programs:

2.1 | TPG

The simulated genome consisted of 10 chromosomes (1.43 Morgans each), 8×10^8 base pairs per chromosome, 2×10^{-9} mutation rate, 1500 segregation sites, and 25 inbred founders forming a population of 50 crosses with 20 progeny each. The trait structure consisted of one complex trait with 50 quantitative trait loci (QTLs)/chromosome where additive effects were simulated from a normal distribution of the form a~MVN$(0,\sigma^2{}_a = 1)$ (where MVN is multivariate normal distribution) and dominance degrees for the QTLs were simulated from a normal distribution of the form δ~MVN$(0,\sigma^2{}_\delta = 0.1)$ to compute dominance effects as $d = \delta|a|$, and no epistatic variance was assumed. A single nucleotide polymorphism chip with 50 markers per chromosome was generated for downstream analysis. It is important to consider that results will be affected by the specific simulated TPG). Therefore, the data simulated here are just an example to show the value of the proposed objective functions.

2.2 | TPE

Note that 300 environments were simulated with genetic correlations drawn from the distribution $r_G \sim$ MVN(0.3, 0.3), representing a broad and heterogeneous TPE. Phenotypes within each environment were simulated to follow the form:

$$y_{ij} = g_i + s_j + e_{ij},$$

where y_{ij} is the phenotype for the jth observation of the ith genotype, g_i is the genetic value from the ith genotype coming from the infinitesimal model $g = \Sigma\alpha + \Sigma\delta$ (average effects of an allele substitution and dominance deviation effects from the QTLs), s_j is the spatial effect associated to the plot following a linear pattern with a distribution s~MVN$(0,\sigma^2{}_s = 50\sigma^2{}_g)$, and e_{ij} is the random error associated to observation y_{ij} with a distribution e~MVN$(0,\sigma^2{}_e = 5\sigma^2{}_g)$. This framework produced plot-level phenotypes with approximately 0.1 reliability, capturing strong spatial variation typical of field trials. It is important to consider that results will be specific to the simulated TPE. The simulated data serve only as a conceptual scaffold to evaluate the proposed objective functions.

2.3 | How many environments represent our TPE, and what level of sparsity represents our TPG?

To address this question, we constructed a three-way grid of treatments varying:

1. ia: unique individuals available for testing/sampling from the entire TPG
2. ipf: number of unique individuals tested per environment/location
3. noe: number of environments available for testing

The product of ipf and noe gives the total number of plots used (pt). In the simulated TPE and TPG, the values of the grid were the combinations of:

$$\text{ia} \in \{100,\ 150,\ 200,\ 1000\}$$

$$\text{ipf} \in \{100,\ 150,\ 200,\ 1000\}$$

$$\text{noe} \in \{3,\ 5,\ 10,\ 15,\ 20,\ 25\}$$

while constraining total plots to pt = 5000. For example:

Individuals available for (ia)	Individuals tested per environment (ipf)	Number of environments available (noe)	Total plots used
1000	100	3	300
200	100	3	300

The scenario in Row 1 indicates that we can sample seed packets from the 1000 unique individuals (ia) forming the TPG, but we can only test up to 100 individuals in each environment (ipf), and since we have three environments available for testing, a maximum of 300 plots and unique individuals may actually be tested. The scenario in Row 2, on the other hand, uses similar levels of resources but is only allowed to sample a smaller set of the TPG (200 out of the possible 1000).

All the combinations of treatments that exceeded the total number of plots of 5000 were removed from the simulation. The best treatment was considered the one with the highest genetic gain and accuracy within the constraints of the total number of plots (pt). Environments and individuals were sampled randomly, with 100 independent replicates, to identify general trends and quantify uncertainty, although in practice some of the proposed objective functions could be used for these simulations to derive more treatment-specific numbers. The equivalent levels of sparsity were recorded and displayed in the associated plots. The level of sparsity is measured with respect to a fully balanced design but always constrained by the maximum number of plots available.

2.4 | Which environments represent best the TPE and should be prioritized?

We evaluated four strategies for selecting the most representative environments:

1. A set of n environments are selected based on *k*-means (we used the *kmeans*() function in base-R with default values) where the number of centers is equal to the number of environments to represent the TPE and the numeric matrix of data is the table of genetic values for each individual in the different environments.
2. A set of n environments is selected based on hierarchical clustering (using the *hclust*() function in the stats package in R, where *hclust* hierarchical clustering), where the distance matrix is based on the table of genetic values for each individual in the different environments and the tree is cut at the height of *n* branches.
3. A set of n environments are selected using the genetic algorithm with objective function $f(\mathbf{q}) = -\mathbf{q}'D\mathbf{q}$ where $\mathbf{q}$ is the contribution vector (0 for non-selected environments, and 1 for selected environments) and D is the relationship matrix between environments (E) computed as the covariance between yield data points (assumed to be known), then we let the evolutionary algorithm run for 50 generations with the defaults used by the *evolafit*() function from the *evola* package (Covarrubias-Pazaran, 2024).
4. A set of *n* environments is selected randomly (control treatment).

All treatments were evaluated in a grid of number of environments, noe = {4, 6, 8, 10, 12, 14, 16, 18, 20, 25} to understand how the number of environments tested/sampled in the phenotypic evaluation affects the response variables (see section 2.8). Later in the discussion, we go over alternative ways to obtain the relationship matrix between environments (E matrix).

2.5 | How to spread entries across environments?

With fixed number of environments of 6 or 25, identified from the previous question, we devised four treatments to decide how to spread a fixed number of 1000 entries across the selected environments with the constraint of a maximum number of plots (ipf by noe product) equal to 1000 and 5000, respectively. This represents some common levels of resources (in terms of number of plots) for our breeding programs when resources are either low or high in a given year.

Treatments:

1. Adverse control: all family members assigned to the same environment
2. Balanced allocation
3. Genetic algorithm optimizing $f(\mathbf{q}) = -\mathbf{q}'D\mathbf{q}$, where D is the Kronecker product of G and E
4. Random sparse allocation (control)

2.6 | How big should the trials be within an environment?

To evaluate trial size and replication, we used grids:

1. Field size: 100–900 plots in increments of 100
2. Replication levels: 1.0, 1.2, 1.5, 1.8, and 2.0

Fields were arranged as square as possible. The different allocations were randomized and analyzed using a mixed model to remove the spatial effects and dissect the genetic signal (see Section 2.10).

2.7 | How to spread entries within an environment?

With the number of environments, specific environments, and allocation of entries across the environments defined, we devised four treatments to identify the most efficient way to spread entries spatially within a field under augmented ($p = 1$) and partially replicated (p-rep) ($p = 1.2$) designs: (A) adverse control with entries from the same family planted in the same portion in the field; (B) baseline of random allocation of entries across the field using a completely randomized design; (C) A-optimality criterion implemented in the odw R package (Cullis et al., 2026) with random formula ~ vm(genotype, Gi) + Row + Range, permute formula ~ vm(genotype, Ai), and residual formula ~units; (D) a genetic-by-spatial objective function (see section 2.9). All designs included 5–10 checks positioned in a diagonal arrangement within the field.

2.8 | Metrics recorded

Across all treatments, we recorded:

1. Accuracy (correlation between true and estimated breeding values across the entire TPE)
2. Expected genetic gain (selecting the top 10%)
3. Computational time
4. A-optimality value (when applicable)

We emphasize that "true breeding value across the TPE" refers to the average of environment-specific true breeding values under an unstructured G × E model.

2.9 | Objective functions to optimize decisions

We used different objective functions to optimize a design depending on the question addressed and these will be explicitly called in the different sections below.

2.9.1 | A-optimality criterion

A-optimality seeks to minimize the average pairwise variance of elementary treatment contrasts. A trials' design is said to be "A-optimal" if its A-objective value is the minimum of the set of A-values across all permutations of the design, assuming that all treatment comparisons are of equal interest (Cullis et al., 2006). The objective function is expressed as:

$$\mathcal{A} = \frac{2}{v-1}\left(tr(\Lambda) - \frac{1}{v}\mathbf{1}_v'\Lambda\mathbf{1}_v\right).$$

where Λ is the inverse of the coefficient matrix of the mixed model equations for the assumed random and residual models and their respective variance-covariance parameters.

Connectedness aims to maximize the inferential information gained from scarce treatment allocation. This is based on the principle of *connectedness* through genetic-by-spatial effects, developed by the authors of this paper. Put less formally, we seek to provide an answer to the following fundamental question (in the form of an optimal solution): How can we optimize the allocation of our treatment entries, across the fields and onto the fields, in such a way that as many accurate inferences as possible can be made regarding how any given treatment entry that was not assigned to a field (this is of course due to limited resources available) would hypothetically develop itself on that field? We do this by considering all inter-genetic relationships between any two treatment entries and by considering all inter-environmental relationships between two fields. These relationships are assumed to be known a priori in our model.

This principle of connectedness makes use of two objective functions in our optimization model, one to distribute the treatment entries to the various fields and one to distribute the treatment entries onto the field plots. Let S represent the number of treatment entries, and F the number of fields.

Then, the first objective function can be formulated as

$$C\ (2) = 1_F^{\mathrm{T}}(E^{\beta}\, Y^{\mathrm{T}} G^{\alpha})1_S,$$

where 1_F and 1_S are summation vector matrices (i.e., vectors of ones) with dimensions $[F \times 1]$ and $[S \times 1]$, respectively (with 1_F^T being the transpose of 1_F), Y is an $[S \times F]$ matrix of decision variables (i.e., $Y_{tf} = 1$ if Treatment Entry t is assigned to Field f, and $Y_{tf} = 0$ otherwise), E is an $[F \times F]$ matrix of inter-fields environmental relationships, and G is an $[S \times S]$ matrix of inter-treatments genetic relationships (genomic relationship matrix [GRM]; VanRaden, 2008). In addition, $\beta \geq 1$ and $\alpha \geq 1$ are calibration parameters used to emphasize the relative importance of these two components (respectively). Meanwhile, let P_f represents the number of plots in Field f.

The second objective function can be formulated as:

$$C\ (3) = 1_{P_f}^{\mathrm{T}}\left[\left(1 - R_f\right) Z_f^{\mathrm{T}} G^{\alpha}\right] 1_S,$$

where 1_{P_f} is a $[P_f \times 1]$ summation vector matrix, $\boldsymbol{1}$ is a $[P_f \times P_f]$ summation matrix, Z is an $[S \times P_f]$ matrix of decision variables (i.e., $Z_{tpf} = 1$ if Treatment Entry t is assigned to Plot p in Field f, and $Z_{tpf} = 0$ otherwise), and $\boldsymbol{R_f}$ is a $[P_f \times P_f]$ matrix of the inter-plots relationship in Field f, whose entries are derived from a 4D mapping function of the inter-rows, inter-ranges, inter-blocks, and spatial distance relationships within that field (details of which can be provided as complementary documentation on request).

An additional function named *maximum representation* objective function is based on optimal contribution theory (Woolliams et al., 2015) modified to only consider the group relationship:

$$f\ (\boldsymbol{q}) = -\boldsymbol{q}' D \boldsymbol{q},$$

where **q** represents the contribution vector with dimensions $e \times 1$, with e being the number of environments available filled with 1s if environment is selected or 0s otherwise. D represents the relationship matrix between environments (E) or between individuals (G), or the Kronecker product between the two depending on the problem being optimized. This objective function was used to find the most representative environments from the TPE and how to allocate the entries in the decided environments (sparse testing).

2.10 | Statistical analysis

In all treatments compared, we required the estimation of breeding values. Depending on the question and the complexity of the model, we used the machinery incorporated in the R packages bWGR (Bayesian Whole Genome Regression) (Xavier et al., 2020), sommer (Covarrubias-Pazaran, 2016), or asreml-R (D. G. Butler et al., 2017). The general mixed model used the following form:

$$\boldsymbol{y} = X\boldsymbol{\beta} + Z\boldsymbol{u} + \boldsymbol{e}$$

where **y** is the vector for the response variable, and X and Z are the incidence matrices linking random effects to the response variable. The vectors $\boldsymbol{\beta}$ and **u** are the fixed and random effects, and **e** is the vector of residuals. The random effects included in the analysis were experimental design factors such as row, range, blocks, a spatial two-dimensional spline kernel, and the genotype effect. Except for genotype, all the other random effects followed a normal distribution with an identity matrix for relationship between levels; $r \sim N(0, \mathrm{I}\ \sigma^2{}_r)$. The genotype effect followed a normal distribution with the GRM to denote relationship between the levels of the factor variable; $\mathbf{g} \sim N(0, \mathrm{G}\ \sigma^2{}_g)$. For multi-environmental data, only the main effect model was used to extract breeding values across the TPE and calculate accuracies across the TPE (no environment-specific accuracies were calculated).

3 | RESULTS

3.1 | How many environments represent our TPE, and what level of sparsity represents our TPG?

We found that for the simulated TPE [$r_G \sim$ MVN(0.3, 0.3)] under the constraint of 5000 total plots, the treatments involving 20 or 25 environments—with 200 individuals tested per environment and covering all 1000 individuals in the TPG—yielded the highest across-TPE accuracies among sparse-testing scenarios (0.48 ± 0.03 for both) (Figure 2). These designs required approximately 80% fewer plot-level observations than a fully balanced design at the same number of environments, yet accuracy declined by only about 20%. Both treatments (20envs/1000ia/200ipf/4000pt and 25envs/1000ia/200ipf/5000pt) exceeded the accuracy of a classical fully balanced design using five environments and 5000 plots (0.45 ± 0.03). Notably, the 20-environment scenario required only 4000 plots to outperform this benchmark. In general, sparse testing scenarios that included the entire TPG across environments produced lower standard errors than scenarios testing only a subset of individuals. Although the latter had higher pairwise connectivity, the broader sampling of individuals resulted in more stable accuracy estimates. Consistently, accuracy increased as the number of environments increased.

We also observed that less-balanced (sparser) testing scenarios performed worse than more-balanced sparse testing scenarios when the number of environments was small (<10). For example, for three environments and using the same number of plots (600), designs testing only 200 unique individuals performed better than designs testing all available 1000 individuals. When the number of environments was small, maximizing per-environment connectivity had greater influence than sampling more individuals. However, as the number of environments increased, the trend reversed: at larger environmental representation (e.g., 20 environments), the less-balanced (sparse) designs outperformed the more-balanced designs (Figure 2). In addition, we found that sampling fewer individuals per environment but across more environments yielded higher accuracies and genetic gains than sampling more individuals in fewer environments.

3.2 | Which environments represent the TPE better?

Once we identified a sparse testing scenario that maximizes the accuracy and genetic gain for the TPE and TPG (and the associated constraint in number of plots), we tested different algorithms to pick the most representative "*n*" environments. We found that both hierarchical clustering (limited to n branches) and *K*-means clustering (with *n* centers) performed similarly to random selection when *n* exceeded six environments, with no meaningful advantage in either accuracy or uncertainty (Figure 3). On average, the modified optimal contribution function approach improved accuracy by approximately 20% relative to the alternative methods and produced smaller standard errors across the full grid of *n* values (Figure 3).

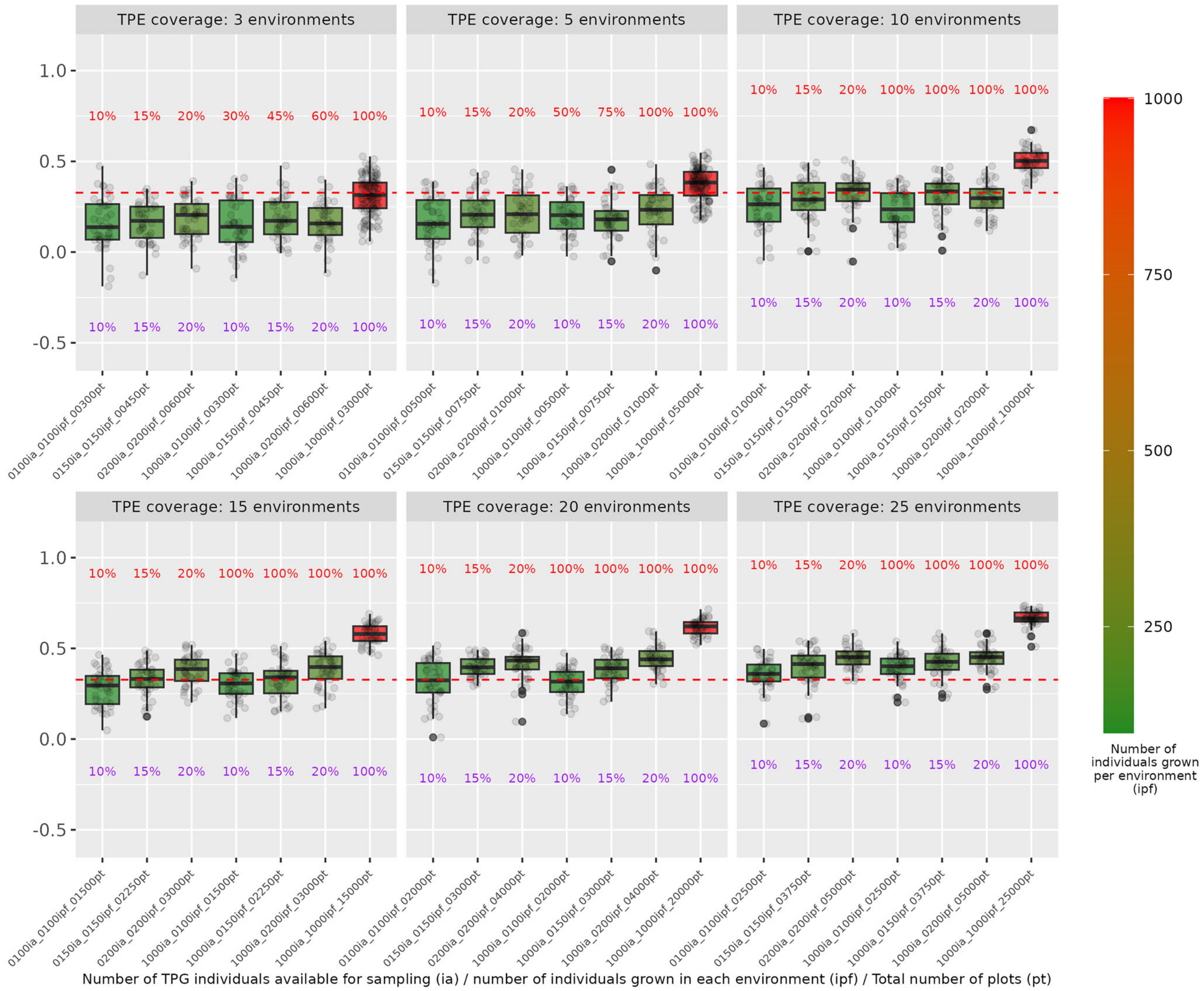

FIGURE 2 Effect in accuracy for a given target population of environments (TPE) and target population of genotypes (TPG). Each treatment (x-axis label in a box) represents three interacting factors: (1a) number of individuals available for testing (ia) *across* the experiment; (2) number of individuals tested per location/environment (ipf); and (3) number of environments (noe; box). Total number of plots planted (pt; noe by ipf) is shown at the end of the treatment label. The color gradient and the percentage labels at the top of boxplots within a box show the level of sparsity of the treatments (green is more spare and red is fully balanced). The label at the bottom of the boxplots within the box represents the percentage of plots used with respect to a fully balanced design. The red dotted line represents the average accuracy across all treatments. Each dot in a boxplot represents an independent simulation. The individuals available (ia) for testing does not imply all of them were actually tested but implies that they were available for the allocation exercise (stronger or weaker population structure in the genetic evaluation).

The pattern for expected genetic gain mirrored the accuracy results. For example, when selecting 25 environments:

1. Hierarchical clustering: accuracy $= 0.55 \pm 0.018$
2. *K*-means: 0.52 ± 0.019
3. Random selection: 0.55 ± 0.017
4. Optimal-contribution ($-\mathbf{q}'\mathrm{D}\mathbf{q}$): 0.60 ± 0.014

Correspondingly, expected genetic gain increased by 8.4% under the use of the modified optimal-contribution function ($\Delta g = 5.27 \pm 0.25$) compared with random selection ($\Delta g = 4.86 \pm 0.27$) and by 10%–15% relative to hierarchical clustering or *K*-means. These improvements were especially pronounced when selecting smaller sets of environments, where representativeness had a greater influence on overall accuracy. The less environments available, the greater the advantage was found of the modified optimal contribution function over clustering and random allocation methods.

Although 20–25 environments had previously been identified as optimal for our resource constraints, we evaluated a full range of n values to characterize how each selection method behaves across both small and large environment subsets.

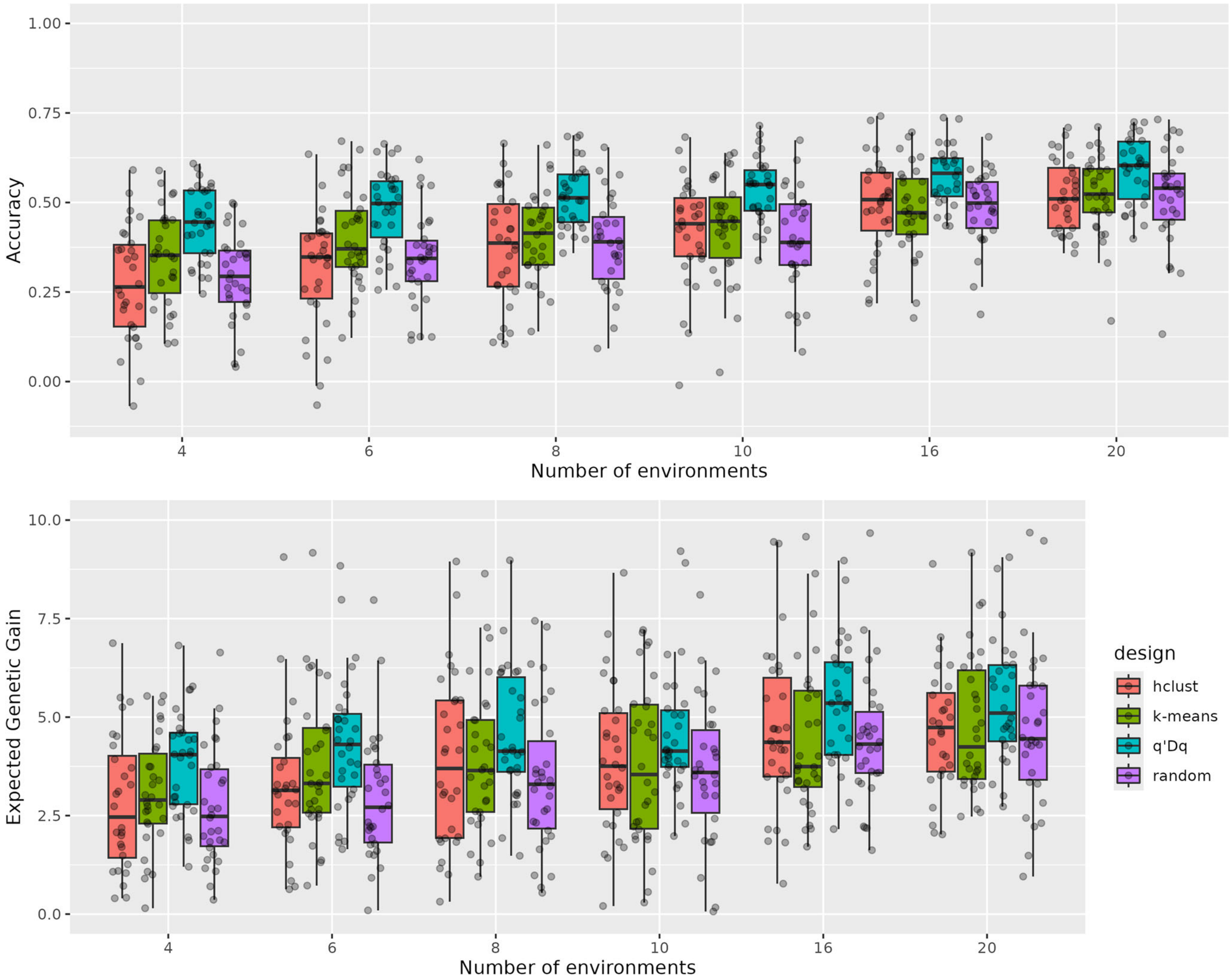

FIGURE 3 Effect in accuracy (upper panel) and genetic gain (lower panel) in a single generation of using different objective functions to pick the 4–20 (increments of 2) most representative environments from a simulated TPE [$r_G \sim$ MVN(0.3,0.3)]. Correlations were calculated between true simulated breeding values and estimated breeding values (accuracy) from a main effect mixed model with the genomic relationship matrix (G). Treatments are as follows: (a) Hierarchical clustering (*hclust*; red); (b) K-mean (*kmeans*; green); (c) Modified optimal contribution function [f(q) = −q′Dq; blue]; (d) Random selection of environments (random; here it represents our control; purple). Each dot in the boxplot represents the result from an independent simulation.

3.3 | How to spread entries across environments?

With the assumption of 20 selected environments, we proceeded to evaluate different objective functions to spread the entries and understand the advantage of a proper allocation of entries across the selected environments. A fully balanced design, in which all entries were tested in all environments, served as a positive control. This design produced the highest accuracy (r = 0.68 ± 0.03) and the lowest standard error. In contrast, systematically spreading/clustering individuals from the same family in a single environment (adverse control) decreased the accuracy of the genetic evaluation while doubling the standard error compared to the positive control (r = 0.39 ± 0.06). The modified optimal contribution function −q′Dq (gafree) achieved an accuracy of r = 0.54 ± 0.04 (Figure 4), outperforming random allocation and substantially outperforming the adverse control, though it remained below the fully balanced benchmark. Repeating the comparison with only five environments yielded the same ranking of treatments, but the differences among methods were smaller, reflecting the reduced complexity and narrower G × E structure associated with fewer environments.

3.4 | How big trials should be within an environment?

We next evaluated how field size and replication level influence accuracy and expected genetic gain within a sin-

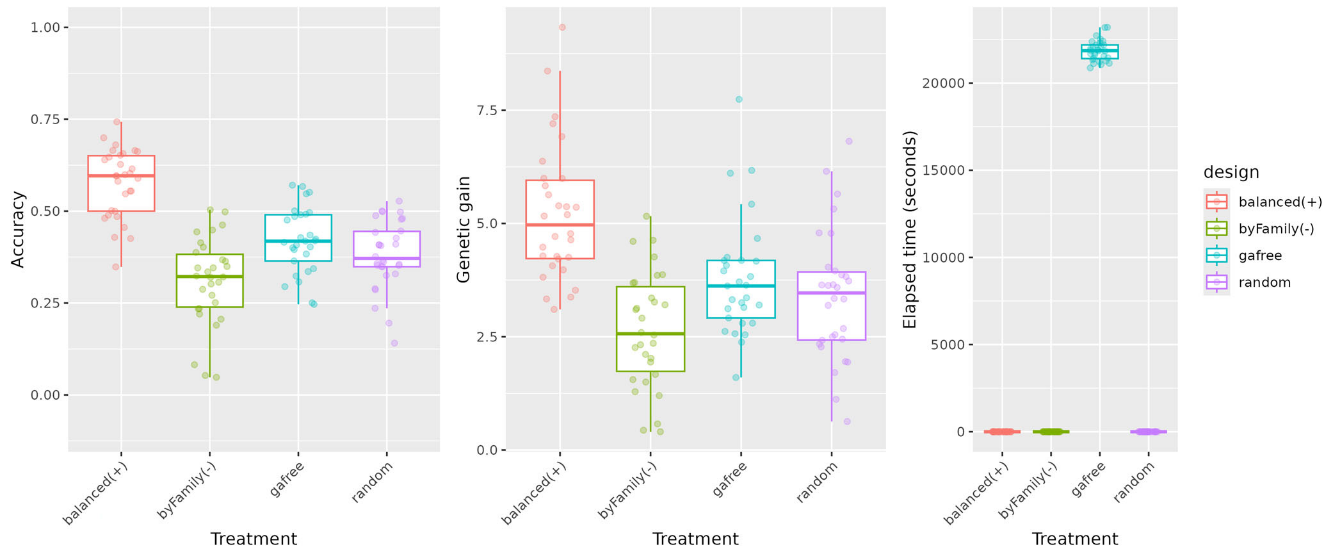

FIGURE 4 Effect in accuracy (left panel), genetic gain (center panel), and computational time (right panel) of using different objective functions to allocate entries in a fixed set of 20 environments, 1000 individuals available for randomization, and a maximum of 5000 plots (200 plots per environment). Treatments (colored boxes on *x*-axis) are as follows: (a) positive baseline (gold standard) where all entries are tested in all selected environments [balanced(+)]; (b) adverse control of sparse testing distribution where entire families go to the same environment [byFamily(−)]; (c) modified optimal contribution function (−q′Dq) using the genetic algorithm (gafree), being D the genetic correlation matrix between environments; (d) random allocation of entries allocated sparsely in the 20 environments. Correlations were calculated between true simulated breeding values and estimated breeding values from a main effect mixed model that includes the genomic relationship matrix (genomic best linear unbiased prediction [GBLUP]). Each dot in the boxplot represents an independent simulation.

gle environment. Trial dimensions ranged from 100 to 900 plots and replication levels from 1.0 to 2.0. Accuracy was calculated across the full TPG, including both tested and fully predicted individuals, allowing evaluation of how each design affected the overall predictive quality.

Under the assumed spatial variation and TPG structure, we found that accuracy increased consistently as field size increased, while the standard error of accuracy decreased, indicating more stable prediction. A plateau emerged at approximately 400–500 plots, beyond which additional increases in field size yielded diminishing returns in accuracy. For example, accuracy improved from ~0.23 with 100 plots (replication level 1) to ~0.44 at 200 plots, then to ~0.60 at 400 plots. Increasing to 800 plots produced a smaller additional gain, reaching ~0.70 (Figure 5).

Differences between replication levels were present but generally smaller than differences attributable to field size. Replication levels influenced accuracy, but their effect was modest relative to the strong influence of total plot number.

Overall, designs with lower replication—allowing more unique entries to be evaluated—tended to yield slightly higher accuracies and genetic gains, although the magnitude of the difference was limited. Trends in expected genetic gain were similar to those observed for accuracy, reflecting the similar underlying statistical behavior.

3.5 | How to spread entries within an environment?

To evaluate strategies for spatially allocating entries within a field, we compared classical and optimized design approaches under both augmented (p = 1) and p-rep (p = 1.2) scenarios. Designs were evaluated for trials containing 200 or 1000 entries, each including 5–10 checks diagonally positioned. diagonally.

We compared four allocation strategies:

1. an adverse control with family members clustered together,
2. a classical incomplete-block randomization,
3. an A-optimality design (odw), and
4. a spatial-by-genetic objective-function design (GbySpat).

In augmented designs with 200 entries, both the A-optimality approach (r = 0.56 ± 0.04) and the GbySpat method (r = 0.54 ± 0.04) outperformed the classical randomization (r = 0.50 ± 0.04). All three performed substantially better than the adverse family-clustering control (r = 0.40 ± 0.06). The A-optimality values corresponded closely with realized accuracy (Figure 6), confirming their usefulness as a model-based proxy for design quality. These performance patterns were consistent across both augmented and p-rep designs and across trials with either 200 or 1000 entries (Figure 6). Computational time differed markedly

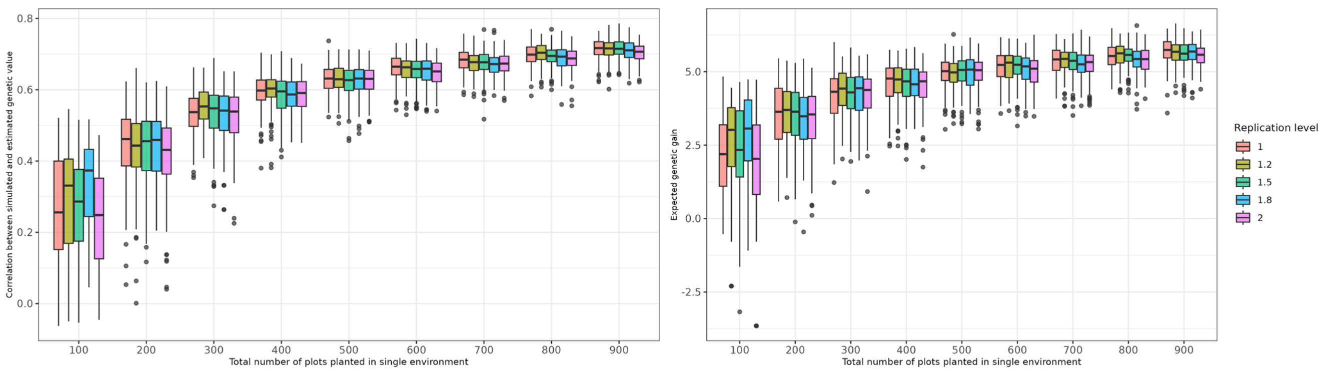

FIGURE 5 Effect in accuracy (left box) and expected genetic gain (right box) is displayed on the *y*-axis as a function of using different field dimensions expressed in number of plots (*x*-axis) to select accurately a target population of genotypes ($N = 1000$) under different replication levels (colored boxplots): unreplicated (1), partially replicated (1.2, 1.5, 1.8), and fully replicated (2). The correlation between true and estimated breeding values was calculated using all the population independently of how many individuals were tested (e.g., in trials with 100 plots, the non-tested entries were predicted to be included in the correlation).

across methods. For trials with 1000 entries, classical randomization required only ~10 ± 2 s, whereas GbySpat required ~1686 ± 13 s (≈28 min), and A-optimality (odw) required ~3346 ± 144 s (≈56 min) on an 8-core Linux system with 5 GB random access memory.

These results highlight that while optimized spatial designs improve accuracy, they come with substantial computational cost—especially for large trials.

4 | DISCUSSION

Although the results of the objective functions applied to different decision points are discussed in the context of resources of a private-sector breeding program, the framework is equally relevant for small and medium-sized public breeding programs. Many of the trends observed reflect resource constraints common across both private and public sectors.

4.1 | How many environments represent our TPE, and what level of sparsity represents our TPG?

The use of sparse-testing designs across environments has taken more relevance in the past years given the advantages to decrease costs of evaluation while suffering small penalties in accuracy (Mothukuri et al., 2025; Atanda et al., 2021) or the ability to test more individuals with the same or similar levels of resources (Garcia-Abadillo et al., 2024; Jarquin et al., 2020). Our simulations likewise showed that sparse-testing strategies performed increasingly well as the number of environments required to represent the TPE increased. Although sparse designs are inherently less informative than fully balanced trials, they become attractive when balanced designs are infeasible due to resource limitations. Under the constraint of 5000 plots, the largest balanced design included 5 environments, whereas sparse-testing approaches enabled coverage of up to 25 environments by distributing smaller trials across locations. Sparse strategies only outperformed balanced designs when at least 15 environments were sampled and consistently surpassed the classical balanced scenario with 20 environments or more under the assumed TPE complexity (Figure 2).

It is acknowledged that comparing scenarios with constant similar number of plots while increasing the number of locations is not straightforward in practice since there are higher costs associated with planting and harvesting more locations. However, our purpose was to demonstrate that, for a TPE of a given complexity [here $r_G \sim MVN(0.3, 0.3)$], a sparse-testing configuration exists that maximizes accuracy relative to a balanced design of the same plot size. Optimal configurations will vary depending on the genetic correlation structure of each program's specific TPE (Cooper et al., 2023). Thus, programs must customize their sparse-testing strategies to their own TPE and TPG characteristics. Better-defined TPEs (narrower distribution in genetic correlations) will decrease the effectiveness of sparse testing approaches, whereas broader TPEs (wider distribution in genetic correlations) will increase the effectiveness of sparse testing scenarios.

Alternatively, sparse testing can be viewed as a strategy to reduce cost while tolerating only a modest loss in accuracy. For example, in Figure 2 we showed how the accuracy of testing at five fully balanced environments versus at five environments with only 20% of the population tested (five times smaller experiments) changed from $r = 0.37$ (non-sparse) to $r = 0.26$ (80% sparse). If evaluation costs scales approximately with trial size, this corresponds to an 80% reduction in cost for a 30% reduction in accuracy. These tradeoffs have also been reported in empirical studies (Jarquin et al., 2020;

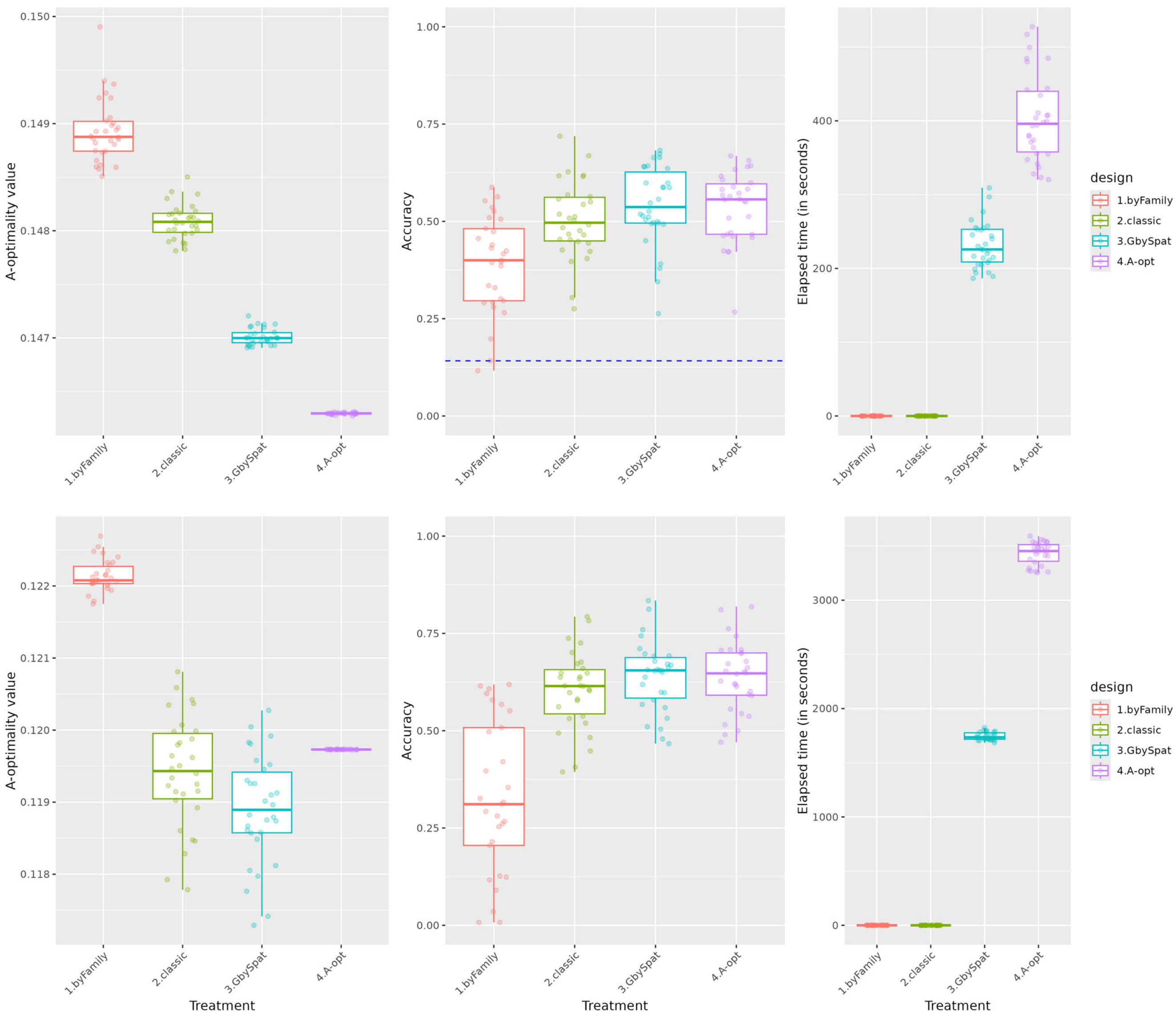

FIGURE 6 A-optimality value (left panels), accuracy (middle panels), and computation time (right panels) between different optimization methods (colored boxes on *x*-axis) in simulated augmented designs using 200 entries (upper row) and 1000 entries (lower row). Treatments are as follows: (a) adverse control allocating family members to the same part of the field; (b) control baseline where entries are allocated following rules of classical experimental design (incomplete block design); (c) genetic by spatial distance algorithm where similar family members are spread across the field (GbySpat); and (d) A-optimality criterion (odw). Each dot in a boxplot represents an independent simulation. In these scenarios the spatial effects are simulated to be strong (see Section 2).

Montesinos-Lopez et al., 2023). The best trade-off needs to be defined by the breeding program based not solely on their resource availability but on the required products in customer's fields.

Breeding programs can rapidly evaluate through simulation similar sparse-testing grids for their own TPE characteristics by introducing in the simulation their genetic correlation structures computed using multi-environmental trial datasets. Such simulations naturally incorporate program-specific G × E behavior, including year-to-year variation, which is often larger than location-specific effects (Atlin et al., 2000).

It is important to note that answers reached by the objective functions applied to the simulated data are specific to the TPE and TPG simulated (e.g., following specific distributions). The simulated data are just an entry point to show the value of the objective functions proposed and how it could be applied by breeding programs to assess allocations. In addition, what we call here "true breeding value across the TPE" is assumed to be the average of all environment-specific true breeding values since an unstructured G × E model assumes that each genotype has a different breeding value at each environment (De Resende & Thompson, 2004). From our perspective, this is a valid assumption, but caution is advised when drawing conclusions for real biological systems.

4.2 | Which environments represent better the TPE?

Selecting a representative subset of environments is a common but often informal step in artificial selection programs. Although breeding programs often rely on long-standing testing locations, certain stages require selecting a smaller—but representative—subset of environments. Our simulations assumed that the environment--relationship matrix was available. Historically, constructing such matrices requires extensive multi-year phenotypic datasets. Today, environmental relationship matrices can be derived using enviromic data from weather, soil, and remote-sensing databases (Costa-Neto, Crossa, et al., 2021; Costa-Neto, Galli, et al., 2021; Fritsche-Neto, 2024). This assumes that all traits would follow the same environmental relationship, which a priori is known to be false, but it can at least serve as a general relationship, and with time the trait-specific G × Es can be linked to specific environmental features to refine the used relationship matrix in the optimization framework proposed (Tolhurst et al., 2022). In addition, the use of envirotyping data allows the inclusion of unseen environments that are of interest for predictive-breeding programs (Jarquin et al., 2020).

Interestingly, classical clustering methods (*K*-means, hierarchical clustering) did not outperform random selection when the number of environments increased to realistic levels of testing. We attribute this to the fact that clustering methods aim to partition environments into groups rather than maximizing the representativeness of the selected subset. In contrast, the modified optimal-contribution function $\mathrm{f}(\mathbf{q}) = -\mathbf{q}'\mathrm{D}\mathbf{q}$ successfully selected environments that increased accuracy and reduced uncertainty. Although originally developed to manage genetic contributions in breeding populations (Woolliams et al., 2015), its structure naturally extends to minimizing redundancy among selected environments and thereby maximizing coverage of the TPE. In practice, identifying the most representative locations requires a proper understanding of the goals of a breeding program, which are (1) improving populations and (2) delivering products in a market segment. The identification of core locations should always be aligned with well-defined segments or mega-environments in some cases. In a side note, selecting high-heritability locations remains useful for ensuring trial quality (Bernardo, 2002) but should only complement—not replace—selection based on representativeness. High-heritability sites may not capture the full diversity of TPE-level G × E patterns.

4.3 | How to spread entries across environments?

The sparse testing methodology is a well-established methodology in plant breeding to optimize the allocation of resources when a fully balanced design is prohibitive but there is the need or interest to cover a representative portion of the TPE (Jarquin et al., 2020; Lubanga et al., 2025; Montesinos-López et al., 2023). Previous studies have described trade-offs between sparse-testing intensity and prediction accuracy (Endelman et al., 2014). On the other hand, our intention was to test what is the best algorithmic way to conduct the spread of the entries is. For that, we tested different treatments that encapsulate different objective functions to spread entries across locations to maximize accuracy and gain. While some would expect to see a big impact of the way we spread entries across the locations and room for improvement, we found that such a situation only occurs when the variance in genetic correlations between the environments is high (mixture of positive and negative correlations) and the number of environments needed to represent the TPE is big.

We found that allocation strategies influenced accuracy mainly when the TPE exhibited large variability in genetic correlations and when many environments were needed to represent it. When few environments were sampled (e.g., 5), minimizing group relationship using $\mathrm{f}(\mathbf{q}) = -\mathbf{q}'\mathrm{D}\mathbf{q}$ did not substantially improve accuracy relative to random allocation. However, under broader TPEs requiring 20 environments, the same approach produced clear improvements. As expected, clustering relatives into the same environment consistently reduced accuracy and increased uncertainty, underscoring the importance of avoiding this practice. While in this research we have focused on the general idea of proposing an objective function to conduct the sparse testing allocation at any given stage, we consider it important to remind the reader that artificial selection programs capitalize the G × E in different ways, either by selecting across the entire TPE or in specific regions (Bernardo, 2002; Cooper et al., 2023), and we expect our objective function to work well under both scenarios.

4.4 | How big should the trials be within an environment?

Field-size and replication analyses confirmed previous findings (Hoefler et al., 2020) while extending them to a broader range of experimental sizes. Our approach compared replication levels within a constant field size, avoiding confounding between trial size and replication intensity. Under high spatial variance (~50 σ^2g), at least 400 plots were needed to sufficiently remove spatial noise, aligning with expectations for spatially heterogeneous fields (Cullis & Gleeson, 1991;

Gilmour et al., 1997; Rodriguez-Alvarez et al., 2018). Testing more unique entries with lower replication was slightly more beneficial than testing fewer entries with more replication when the GRM was used. Models to remove spatial variation can be found elsewhere (Cullis & Gleeson, 1991; Gilmour et al., 1997; Rodriguez-Alvarez et al., 2018). However, differences were modest once a minimum number of entries were tested, implying that replication level is of secondary importance when GRM-based models are used (Figure 5). In practice, each program would need to run these simulations (which only take few minutes) to understand the expected results for their specific levels of spatial variation.

4.5 | How to spread entries within an environment?

The spatial allocation of entries remains a major focus in experimental design research (Hoshmand, 2018; Mead, 2017; Petersen, 1994). Our intention was not to compare all possible designs but rather to illustrate the value of optimization-based allocations relative to classical randomization and to quantify the cost of poor allocation choices using objective functions. For example, some programs tend to not randomize their entries but to put families close together for practical purposes, which we labeled in Figure 6 as a "byFamily" design. We found that such designs have much lower accuracies and genetic gains than their counterparts such as randomized spatial designs and optimized designs. This is a well-known behavior, but we aimed to provide a quantifiable measure of how hurtful this practice can be (about 30% less accurate when the spatial variation is high and a relationship matrix is used, so it can potentially be worse). We also found that optimized designs that look to minimize the prediction error variance (e.g., *odw*) or maximize connectedness (genetic-by-spatial) performed better than classical designs based on traditional randomization methods when the spatial variation is high but did not have an effect in accuracy when the spatial variation is low. It is important to consider that such optimization techniques, while being theoretically optimal, can be computationally intensive and drastically increase the time to fit a design when the number of entries is large (e.g., thousands of entries). For example, under the high spatial variance scenario, the classical design based on randomization was fast and accuracies were almost as good as optimized designs ($r = 0.50 \pm 0.04$), and although optimized designs were 8% (ifd) to 12% (odw) more accurate, they also took magnitudes of time longer to run (few seconds for classical versus minutes [genetic-by-spatial] and hours [A-optimality] for optimized big trials). While we would recommend using objective functions to increase the accuracy, we would also recommend considering the computational time required for it. The heuristics used by the genetic-by-spatial approach were the best in finding a sweet spot between accuracy increase and computational time. This is because the heuristic used in the genetic-by-spatial approach does not require the inverse of the coefficient matrix used by the A-optimality criterion implemented in the popular *odw* software (Cullis et al., 2006).

4.6 | Limitations of the study

While we have shown a noticeable effect on the accuracy of phenotypic evaluations when the proposed objective functions are used, these assume the availability of genomic and environmental relationship matrices. Many real programs may not have it readily available, making the application of the proposed methods limited. For example, environmental relationships based on trait-specific data are treated as stable once they are estimated from a multi-environmental trial analysis, while in practice, trait-specific G × E patterns can differ substantially as data accumulate, and environmental relationships based on climate information may not correspond with trait-specific G × E patterns.

The reader should be aware that findings presented in this study are scenario-specific (a specific TPE and a specific TPG), but we did not fully explore how robust the conclusions are to alternative TPE/TPG architectures, and we assume it extrapolates well to other structures. In addition, the study is based on simulated data (no real or semi-real data examples were used) using the infinitesimal quantitative genetics model and the unstructured G × E model (no main genotype effect), which may miss certain biological aspects or signals. Interpretation of the findings should consider such assumptions, which we consider robust enough to be shared with the community but should serve as a starting point for further research.

5 | CONCLUSION

Experimental design across and within locations is central to reducing noise and improving genetic evaluation in artificial selection programs. Approaching this topic holistically is complex due to the many interacting factors. Here, we addressed this complexity by partitioning the design process into discrete decision points, each optimized with a tailored objective function to approximate near-optimal allocations. Our results highlight consistent interactions among allocation factors, enabling programs to tailor recommendations to their own TPEs and TPGs. It is expected that these two populations vary across programs, and thus recommendations must be customized. Our main proposal is to address key allocation decisions using objective functions grounded in quantitative genetics principles, with representativeness as a unifying concept.

Stochastic simulations demonstrated the value of these functions across a range of scenarios. In across-environment decisions, we found that sampling the TPE effectively—determining the number of environments, the sparsity level, and which environments to include—had a larger impact on accuracy than the specific allocation of entries across environments. Random allocations were often close to optimal once poor designs were avoided.

For within-environment decisions, we found that a minimum field size was required to sufficiently model spatial variation (when using spatial models or GRM-based evaluations). The exact threshold depends on spatial noise and population structure, but it was around 400 plots in our simulations. In addition, replication level had a smaller effect on accuracy once enough entries were tested, although small field sizes exhibited greater uncertainty regardless of replication. Finally, spatial allocation affected accuracy and genetic gain primarily under strong spatial variation. Under such conditions, optimized designs (A-optimality or genetic-by-spatial distance) improved performance relative to standard randomization, though with increased computational cost.

Overall, the proposed decision-point framework and objective functions provide a practical and flexible approach for designing phenotypic evaluations that balance accuracy, resource use, and computational feasibility.

AUTHOR CONTRIBUTIONS

Alexandre Colmant: Conceptualization; formal analysis; investigation; methodology; software; writing—review and editing. **Fabiano Pita**: Conceptualization; methodology; project administration; writing—review and editing. **Giovanny Covarrubias-Pazaran**: Conceptualization; formal analysis; investigation; methodology; software; validation; visualization; writing—original draft; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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SUPPORTING INFORMATION

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